

Supplementary Material for “Magnetization Plateaux as a Roadmap to Quantum Spin Liquids”

Anna Keselman,¹ Xinyuan Xu,² Hao Zhang,³ Cristian D. Batista,^{3,4} and Oleg A. Starykh²

¹*Department of Physics, Technion, Haifa, 3200003, Israel*

²*Department of Physics and Astronomy, University of Utah, Salt Lake City, Utah 84112, USA*

³*Department of Physics and Astronomy, University of Tennessee, Knoxville, TN 37996, USA*

⁴*Neutron Scattering Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, USA*

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I. LINEAR SPIN-WAVE THEORY ANALYSIS OF SUBLATTICE MAGNETIZATIONS IN THE Y STATE

In describing three-sublattice states, we closely follow Ref. [1]. Our LSW Hamiltonian is given by Eq. (22) of this reference above with $b = 0$ in Eqs. (25) and (26). More details are provided below.

For a spin-wave expansion, the laboratory reference frame $\{x_0, y_0, z_0\}$ is rotated to the *local* reference frame $\{x, y, z\}$ on each site, so that the z axis is along the direction dictated by a classical spin configuration obtained. For the coplanar Y and V states in Fig. 4 of Ref. [1], such a transformation is a simple rotation in the x_0 - z_0 plane, such that $S_\alpha^{y_0} = S_\alpha^y$ and

$$\begin{aligned} S_\alpha^{x_0} &= S_\alpha^x \cos \tilde{\alpha} - S_\alpha^z \sin \tilde{\alpha}, \\ S_\alpha^{z_0} &= S_\alpha^z \cos \tilde{\alpha} + S_\alpha^x \sin \tilde{\alpha}, \end{aligned} \quad (\text{S1})$$

where α and $\tilde{\alpha}$ are, respectively, sublattice index and corresponding spin angles in Fig. 4(b) of Ref. [1].

This is followed by the standard Holstein-Primakoff bosonization of spins in the rotated reference frame, $S_i^z = S - a_i^\dagger a_i$ and $S_i^- = a_i^\dagger \sqrt{2S}$.

In the three-sublattice state, the site coordinate i is replaced by the unit cell coordinate ℓ and sublattice index α : $i \rightarrow \{\alpha, \ell\}$. As a result, the Holstein-Primakoff boson operators are split into three species $a_{\alpha, \ell}^{(\dagger)} = \{a_\ell^{(\dagger)}, b_\ell^{(\dagger)}, c_\ell^{(\dagger)}\}$ corresponding to $\alpha = \{A, B, C\}$ sublattices. The Fourier transformation for them is

$$a_{\alpha, \ell} = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{q}} a_{\alpha, \mathbf{q}} e^{-i\mathbf{q} \cdot \mathbf{r}_{\alpha, \ell}}, \quad (\text{S2})$$

where $\mathbf{r}_{\alpha, \ell} = \mathbf{R}_\ell + \boldsymbol{\rho}_\alpha$ and $N_c = N/3$ is the number of unit cells. The sublattice coordinates within the unit cell can be chosen as $\boldsymbol{\rho}_A = 0$, $\boldsymbol{\rho}_B = -\boldsymbol{\delta}_2$, and $\boldsymbol{\rho}_C = \boldsymbol{\delta}_3$, see Fig. 3 of Ref. [1].

Using these boson species and their Fourier transforms, the LSWT Hamiltonian for an arbitrary coplanar three-sublattice state reads as

$$\hat{\mathcal{H}}^{(2)} = \frac{3J_1 S}{2} \sum_{\mathbf{q}} \hat{\mathbf{x}}_{\mathbf{q}}^\dagger \hat{\mathbf{H}}_{\mathbf{q}} \hat{\mathbf{x}}_{\mathbf{q}}, \quad (\text{S3})$$

where $\hat{\mathbf{x}}_{\mathbf{q}}^\dagger = (a_{\mathbf{q}}^\dagger, b_{\mathbf{q}}^\dagger, c_{\mathbf{q}}^\dagger, a_{-\mathbf{q}}, b_{-\mathbf{q}}, c_{-\mathbf{q}})$ and $\hat{\mathbf{H}}_{\mathbf{q}}$ is a matrix

$$\hat{\mathbf{H}}_{\mathbf{q}} = \begin{pmatrix} \hat{\mathbf{A}}_{\mathbf{q}} & \hat{\mathbf{B}}_{\mathbf{q}} \\ \hat{\mathbf{B}}_{\mathbf{q}}^\dagger & \hat{\mathbf{A}}_{-\mathbf{q}}^* \end{pmatrix}, \quad (\text{S4})$$

with the 3×3 matrices $\hat{\mathbf{A}}_{\mathbf{q}}$ and $\hat{\mathbf{B}}_{\mathbf{q}}$

$$\hat{\mathbf{A}}_{\mathbf{q}} = \begin{pmatrix} A_{\mathbf{q}} & D_{\mathbf{q}} & E_{\mathbf{q}}^* \\ D_{\mathbf{q}}^* & B_{\mathbf{q}} & F_{\mathbf{q}} \\ E_{\mathbf{q}} & F_{\mathbf{q}}^* & C_{\mathbf{q}} \end{pmatrix}, \quad \hat{\mathbf{B}}_{\mathbf{q}} = \begin{pmatrix} G & J_{\mathbf{q}} & K_{\mathbf{q}}^* \\ J_{\mathbf{q}}^* & H & L_{\mathbf{q}} \\ K_{\mathbf{q}} & L_{\mathbf{q}}^* & I \end{pmatrix}. \quad (\text{S5})$$

The elements of the $\hat{\mathbf{A}}_{\mathbf{q}}$ matrix are given by

$$\begin{aligned} A_{\mathbf{q}} &= h \cos \beta - \cos \tilde{\alpha}_{AB} - \cos \tilde{\alpha}_{AC} - 2j_2(1 - \gamma_{\mathbf{q}}^{(2)}), \\ B_{\mathbf{q}} &= h \cos \alpha_1 - \cos \tilde{\alpha}_{AB} - \cos \tilde{\alpha}_{BC} - 2j_2(1 - \gamma_{\mathbf{q}}^{(2)}), \\ C_{\mathbf{q}} &= h \cos \alpha_2 - \cos \tilde{\alpha}_{BC} - \cos \tilde{\alpha}_{AC} - 2j_2(1 - \gamma_{\mathbf{q}}^{(2)}), \\ D_{\mathbf{q}} &= \gamma_{\mathbf{q}} \cos^2(\tilde{\alpha}_{AB}/2), \\ E_{\mathbf{q}} &= \gamma_{\mathbf{q}} \cos^2(\tilde{\alpha}_{AC}/2), \\ F_{\mathbf{q}} &= \gamma_{\mathbf{q}} \cos^2(\tilde{\alpha}_{BC}/2), \end{aligned} \quad (\text{S6})$$

and of the $\hat{\mathbf{B}}_{\mathbf{q}}$ matrix, respectively,

$$\begin{aligned} G &= H = I = 0, \\ J_{\mathbf{q}} &= -\gamma_{\mathbf{q}} \sin^2(\tilde{\alpha}_{AB}/2), \\ K_{\mathbf{q}} &= -\gamma_{\mathbf{q}} \sin^2(\tilde{\alpha}_{AC}/2), \\ L_{\mathbf{q}} &= -\gamma_{\mathbf{q}} \sin^2(\tilde{\alpha}_{BC}/2), \end{aligned} \quad (\text{S7})$$

where $h = g\mu_B H/3J_1 S$, $j_2 = J_2/J_1$, $\tilde{\alpha}_{AB} = \beta - \alpha_1$, $\tilde{\alpha}_{AC} = \beta - \alpha_2$, $\tilde{\alpha}_{BC} = \alpha_1 - \alpha_2$, and

$$\gamma_{\mathbf{q}} = \frac{1}{3} \sum_{\alpha} e^{i\mathbf{q} \cdot \boldsymbol{\delta}_{\alpha}}, \quad \gamma_{\mathbf{q}}^{(2)} = \frac{1}{3} \sum_{\alpha} \cos \mathbf{q} \cdot \mathbf{a}_{\alpha}, \quad (\text{S8})$$

with the first- and second-neighbor translation vectors $\boldsymbol{\delta}_1 = (1, 0)a$, $\boldsymbol{\delta}_2 = (-1, \sqrt{3})a/2$, $\boldsymbol{\delta}_3 = -(1, \sqrt{3})a/2$, and $\mathbf{a}_1 = (3, -\sqrt{3})a/2$, $\mathbf{a}_3 = (0, \sqrt{3})a$, $\mathbf{a}_2 = -(3, \sqrt{3})a/2$, respectively, see Fig. 3 of Ref. [1]; a is the lattice constant. The equilibrium angles of the Y state are: $\beta = \pi$, $\alpha_1 = -\alpha_2 = \arccos[(1+h)/2]$.

The next step is to find the coefficients U and V of the generalized Bogolyubov transformation from the Holstein-Primakoff bosons to the quasiparticle eigenmodes

$$a_{\alpha, \mathbf{q}} = \sum_{\gamma} \left(U_{\alpha, \mathbf{q}}^{(\gamma)} A_{\gamma, \mathbf{q}} + V_{\alpha, \mathbf{q}}^{(\gamma)} A_{\gamma, -\mathbf{q}}^\dagger \right), \quad (\text{S9})$$

with the quasiparticle operators $A_{\gamma, \mathbf{q}} = \{A_{\mathbf{q}}, B_{\mathbf{q}}, C_{\mathbf{q}}\}$.

In matrix form, the transformation (S9) is

$$\hat{\mathbf{x}}_{\mathbf{q}} = \begin{pmatrix} \hat{\mathbf{a}}_{\mathbf{q}} \\ \hat{\mathbf{a}}_{-\mathbf{q}}^\dagger \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{U}}_{\mathbf{q}} & \hat{\mathbf{V}}_{\mathbf{q}} \\ \hat{\mathbf{V}}_{-\mathbf{q}}^* & \hat{\mathbf{U}}_{-\mathbf{q}}^* \end{pmatrix} \begin{pmatrix} \hat{\mathcal{A}}_{\mathbf{q}} \\ \hat{\mathcal{A}}_{-\mathbf{q}}^\dagger \end{pmatrix} = \hat{\mathbf{S}}_{\mathbf{q}} \cdot \hat{\mathbf{z}}_{\mathbf{q}}, \quad (\text{S10})$$

where vectors $\hat{\mathbf{a}}_{\mathbf{q}} = [a_{\mathbf{q}}, b_{\mathbf{q}}, c_{\mathbf{q}}]^T$, $\hat{\mathbf{a}}_{-\mathbf{q}}^\dagger = [a_{-\mathbf{q}}^\dagger, b_{-\mathbf{q}}^\dagger, c_{-\mathbf{q}}^\dagger]^T$ and $\hat{\mathcal{A}}_{\mathbf{q}} = [A_{\mathbf{q}}, B_{\mathbf{q}}, C_{\mathbf{q}}]^T$, $\hat{\mathcal{A}}_{-\mathbf{q}}^\dagger = [A_{-\mathbf{q}}^\dagger, B_{-\mathbf{q}}^\dagger, C_{-\mathbf{q}}^\dagger]^T$ are introduced. It follows that the transformation matrix $\hat{\mathbf{S}}_{\mathbf{q}}$ diagonalizes $\hat{\mathbf{g}}\hat{\mathbf{H}}_{\mathbf{q}}$ in (S4), where the “commutation” matrix $\hat{\mathbf{g}}$ is defined by $\hat{\mathbf{g}} = \text{Diag}(1, 1, 1, -1, -1, -1)$, as explained in Ref. [1]. Thus, the $U_{\alpha}^{(\gamma)}$ and $V_{\alpha}^{(\gamma)}$ parameters can be extracted as the elements of the properly normalized eigenvectors of $\hat{\mathbf{g}}\hat{\mathbf{H}}_{\mathbf{q}}$ from a diagonalization procedure.

It is useful to observe that $\hat{\mathbf{A}}_{-\mathbf{q}}^* = \hat{\mathbf{A}}_{\mathbf{q}}$ and $\hat{\mathbf{B}}_{\mathbf{q}}^\dagger = \hat{\mathbf{B}}_{\mathbf{q}}$ in (S4) as their off-diagonal matrix elements (S6), (S7) are simply proportional to the complex hopping amplitude $\gamma_{-\mathbf{q}}^* = \gamma_{\mathbf{q}}$ in Eq. (S8). As a result, all eigenenergies of $\hat{\mathbf{g}}\hat{\mathbf{H}}_{\mathbf{q}}$ are reciprocal, $\omega_{\gamma, -\mathbf{q}} = \omega_{\gamma, \mathbf{q}}$, and $\{\hat{\mathbf{V}}_{-\mathbf{q}}^*, \hat{\mathbf{U}}_{-\mathbf{q}}^*\} = \{\hat{\mathbf{V}}_{\mathbf{q}}, \hat{\mathbf{U}}_{\mathbf{q}}\}$ in (S10).

At this point we observe that the order parameter for the sublattice $\alpha = A, B, C$ is given, for $S = 1/2$, by

$$\langle S_{\mathbf{r}}^z \rangle = \frac{1}{2} - \langle a_{\alpha, \mathbf{r}}^\dagger a_{\alpha, \mathbf{r}} \rangle = \frac{1}{2} - N_c^{-1} \sum_{\mathbf{q} \in \text{MBZ}} \sum_{\gamma=1,2,3} |V_{\alpha, \mathbf{q}}^{(\gamma)}|^2, \quad (\text{S11})$$

where MBZ denotes the magnetic BZ. Importantly, we do not need U, V coefficients per se, only the specific combination $\sum_{\gamma=1,2,3} |V_{\alpha, \mathbf{q}}^{(\gamma)}|^2$.

To evaluate (S11), we reformulate the matrix diagonalization problem as the matrix equation for boson Green's functions along the lines of [2, 3], simplifying it significantly. Specifically, we define retarded Green's functions

$$G_{j1}^{-+}(k, t) = -i\theta(t) \langle [\hat{a}_{jk}(t), \hat{a}_{1k}^\dagger(0)] \rangle, \quad G_{j1}^{++}(k, t) = -i\theta(t) \langle [\hat{a}_{j-k}^\dagger(t), \hat{a}_{1k}^\dagger(0)] \rangle, \quad (\text{S12})$$

where $\hat{a}_{jk}$ (or its conjugate) represents the j -th element of the operator vector $\hat{\mathbf{a}}_{\mathbf{k}}$ (or its conjugate), with $j = 1, 2, 3$. These obey the equation of motion

$$-i\partial_t G_{j1}^{-+}(k, t) = -\delta(t) \langle [\hat{a}_{jk}(0), \hat{a}_{1k}^\dagger(0)] \rangle - i\theta(t) \langle [[\hat{\mathcal{H}}^{(2)}, \hat{a}_{jk}(t)], \hat{a}_{1k}^\dagger(0)] \rangle. \quad (\text{S13})$$

The first term on the right-hand side is just δ_{j1} , of course. Equation of motion for $G_{j1}^{++}(k, t)$ does not have a similar term because the commutator of two creation (or annihilation) operators is zero. Using the explicit form of the Hamiltonian (S3) and Fourier transforming as $G(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t} G(\omega)$, one arrives at the matrix equation

$$(\hat{\mathbf{H}}_{\mathbf{k}} - \omega \hat{\mathbf{g}})_{ji} \begin{pmatrix} G_{i1}^{-+} \\ G_{i1}^{++} \end{pmatrix} = \begin{pmatrix} -1 \\ \mathbf{0}_5 \end{pmatrix}_{j1} \quad (\text{S14})$$

where the Hamiltonian matrix on the left is defined in Eq. (S4). The vector $(-1, \mathbf{0}_5)^T$ on the right-hand side is symbolic notation for $(-1, 0, 0, 0, 0, 0)^T$, while that on the left-hand side is made of the two length-3 vectors of the Green's functions defined above.

This equation can now be solved by Cramer's rule: the matrix equation $\hat{A}\mathbf{x} = \mathbf{b}$, where $\hat{A}$ is an $n \times n$ matrix with a nonzero determinant, is solved by $x_i = \det(\hat{A}_i) / \det(\hat{A})$, where $\hat{A}_i$ is the matrix formed by replacing the i -th column of $\hat{A}$ by the column vector $\mathbf{b}$.

In our case, the determinant in the denominator is just $\det(\hat{\mathbf{H}}_{\mathbf{k}} - \omega \hat{\mathbf{g}}) = \Pi_{n=1}^3 (\epsilon_n(k) - \omega)(\epsilon_n(k) + \omega)$, in terms of magnon eigenvalues $\epsilon_n(k)$. Therefore, the Green's function $G_{11}^{-+}(k, \omega)$ has poles at $\omega = \pm \epsilon_n(k)$. In particular, the residue of the pole at $\omega = \epsilon_n(k)$ is given by

$$|U_{1, \mathbf{k}}^{(n)}|^2 = - \frac{\det(\hat{\mathbf{H}}_{\mathbf{k}} - \epsilon_n(k) \hat{\mathbf{g}})_1}{2\epsilon_n(k) \Pi_{m \neq n} (\epsilon_n(k)^2 - \epsilon_m(k)^2)}, \quad (\text{S15})$$

where the matrix $(\hat{\mathbf{H}}_{\mathbf{k}} - \epsilon_n(k) \hat{\mathbf{g}})_1$ in the numerator is the analogue of $\hat{A}_1$ of the Cramer's rule.

The identification of the residue with $|U_{1, \mathbf{k}}^{(n)}|^2$ follows from applying eigenmode expansion (S9) to $G_{11}^{-+}(k, \omega)$, which results in

$$G_{11}^{-+}(k, \omega) = \sum_{n=1}^3 \left(\frac{|U_{1, \mathbf{k}}^{(n)}|^2}{\omega - \epsilon_n(k) + i0} - \frac{|V_{1, \mathbf{k}}^{(n)}|^2}{\omega + \epsilon_n(-k) + i0} \right). \quad (\text{S16})$$

Matching the pole of this expression at $\omega = \epsilon_n(k)$ to the one obtained from Cramer's rule leads to the identification (S15).

Finally, using the standard $U - V$ identity

$$\sum_{\gamma} \left(|U_{\alpha, \mathbf{q}}^{(\gamma)}|^2 - |V_{\alpha, \mathbf{q}}^{(\gamma)}|^2 \right) = 1, \quad (\text{S17})$$

gives us an algebraic way to evaluate $\sum_{\gamma=1}^3 |V_{\alpha, \mathbf{q}}^{(\gamma)}|^2$ at momentum $\mathbf{q}$ in the magnetic BZ in Eq. (S11). The described manipulations illustrate the case of the A-sublattice, when $\alpha = 1$, resulting in $\langle S_A^z \rangle$. Obviously, its extension to B/C sublattices requires one to analyze Green's functions $G_{j2/3}^{\pm+}$ in a similar way.

The remaining sum over $\mathbf{q}$ is done numerically, using the Gaussian quadrature technique. All of the described matrix manipulations are implemented in the Mathematica program. The results of the calculation are illustrated by the dashed red curves with up/down triangle markers in Fig. 4(b) of the main text.

II. LINEAR SPIN-WAVE THEORY ANALYSIS OF SUBLATTICE MAGNETIZATIONS IN THE STRIPE STATE

Next, we present a single-boson formulation of the stripe phase. This is based on the sublattice-dependent rotation of spins from the laboratory frame (x_0, y_0, z_0) to the local one (x, y, z) . For the spin operator at the site $\mathbf{r}$,

$$S_{\mathbf{r}}^{x_0} = \sin \theta S_{\mathbf{r}}^x + \cos \theta e^{iM_1 \cdot \mathbf{r}} S_{\mathbf{r}}^z, S_{\mathbf{r}}^{y_0} = S_{\mathbf{r}}^y, S_{\mathbf{r}}^{z_0} = -\cos \theta e^{iM_1 \cdot \mathbf{r}} S_{\mathbf{r}}^x + \sin \theta S_{\mathbf{r}}^z. \quad (\text{S18})$$

Here $M_1 = (0, 2\pi/\sqrt{3})$, and the spin-wave expansion in the local frame is $S^z = S - a^+ a$, $S^x = \sqrt{S/2}(a + a^+)$, $S^y = -i\sqrt{S/2}(a - a^+)$. By requiring the linear in magnons a, a^+ terms to vanish, the optimal rotation angle is found to be $\sin \theta = B/(8S(J_1 + J_2))$. The harmonic Hamiltonian of the spin waves in the stripe phase is found to be

$$H_{\text{stripe}}^{(2)} = S \sum_{\mathbf{k} \in \text{BZ}} A_{\mathbf{k}} a_{\mathbf{k}}^+ a_{\mathbf{k}} - \frac{1}{2} B_{\mathbf{k}} (a_{\mathbf{k}} a_{-\mathbf{k}} + \text{h.c.}), \quad (\text{S19})$$

$$A_{\mathbf{k}} = 2J_1 \left[1 + \cos \mathbf{k} \cdot \boldsymbol{\delta}_1 + \sin^2 \theta (\cos \mathbf{k} \cdot \boldsymbol{\delta}_2 + \cos \mathbf{k} \cdot \boldsymbol{\delta}_3) \right] + 2J_2 \left[1 + \cos \mathbf{k} \cdot \mathbf{e}_2 + \sin^2 \theta (\cos \mathbf{k} \cdot \mathbf{e}_1 + \cos \mathbf{k} \cdot \mathbf{e}_3) \right],$$

$$B_{\mathbf{k}} = \cos^2 \theta \left[2J_1 (\cos \mathbf{k} \cdot \boldsymbol{\delta}_2 + \cos \mathbf{k} \cdot \boldsymbol{\delta}_3) + 2J_2 (\cos \mathbf{k} \cdot \mathbf{e}_1 + \cos \mathbf{k} \cdot \mathbf{e}_3) \right]$$

As usual, the magnon dispersion is then $\omega_{\mathbf{k}} = \sqrt{A_{\mathbf{k}}^2 - B_{\mathbf{k}}^2}$, while the ordered moment

$$\langle S^z \rangle = \frac{1}{2} - \frac{1}{N} \sum_{\mathbf{k} \in \text{BZ}} \left(\frac{A_{\mathbf{k}}}{\omega_{\mathbf{k}}} - 1 \right). \quad (\text{S20})$$

The last expression is used to derive the spin-wave boundary of the stripe phase in Fig. 4(b) (dashed red line with square markers). The magnetization and susceptibility, Eq. (4) of the main text/End Matter, are obtained by subsequent differentiating of the ground state energy $E_{\text{gs}} = H_{\text{stripe}}^{(0)} + \frac{1}{2} \sum_{\mathbf{k}} (\omega_{\mathbf{k}} - A_{\mathbf{k}})$ with respect to B . Here, the classical energy $H_{\text{stripe}}^{(0)} = -S^2(J_1 + J_2) - B^2/(16(J_1 + J_2))$.

III. SEMICLASSICAL SPIN-WAVE THEORY ANALYSIS FOR THE MAGNETIZATION PLATEAUX

We first note that the UUD (UUUD) configuration is a classical ground state only at the fine-tuned magnetic field $B_{\text{cl}} = 3J_1 S$ [$B_{\text{cl}} = 4(J_1 + J_2)S$]. Following Refs. [4, 5], all spin-wave calculations within the plateau are therefore initialized at this classical field B_{cl} . In the following, we present the calculation for the UUD order as an illustrative example. The results for both the UUD and UUUD plateaux can be reproduced using the numerical code provided in the accompanying repository [6], which implements the derivations below within the Sunny platform [7, 8].

In the UUD phase, the LSWT Hamiltonian can be written in the Bogoliubov form

$$\mathcal{H}_{\text{LSWT}} = \frac{1}{2} \sum_{\mathbf{q} \in \text{MBZ}} ((\mathbf{a}_{\mathbf{q}}^\dagger)^T \ (\mathbf{a}_{-\mathbf{q}})^T) H_0^{(2)}(\mathbf{q}) \begin{pmatrix} \mathbf{a}_{\mathbf{q}} \\ \mathbf{a}_{-\mathbf{q}}^\dagger \end{pmatrix}, \quad H_0^{(2)}(\mathbf{q}) \equiv \begin{pmatrix} H_{11}^0(\mathbf{q}) & H_{12}^0(\mathbf{q}) \\ H_{21}^0(\mathbf{q}) & H_{22}^0(\mathbf{q}) \end{pmatrix}. \quad (\text{S21})$$

Here the momentum summation is restricted to the magnetic Brillouin zone (MBZ), and

$$\mathbf{a}_{\mathbf{q}} = \begin{pmatrix} a_{1,\mathbf{q}} \\ a_{2,\mathbf{q}} \\ a_{3,\mathbf{q}} \end{pmatrix}, \quad \mathbf{a}_{-\mathbf{q}}^\dagger = \begin{pmatrix} a_{1,-\mathbf{q}}^\dagger \\ a_{2,-\mathbf{q}}^\dagger \\ a_{3,-\mathbf{q}}^\dagger \end{pmatrix} \quad (\text{S22})$$

are the Holstein-Primakoff (H-P) boson spinors associated with the three magnetic sublattices of the UUD state. The block matrices are given by

$$H_{11}^0(\mathbf{q}) = S \begin{pmatrix} BS^{-1} + 6J_2(\gamma_{\mathbf{q}}^{(2)} - 1) & 3J_1\gamma_{\mathbf{q}}^{(1)} & 0 \\ 3J_1\gamma_{-\mathbf{q}}^{(1)} & BS^{-1} + 6J_2(\gamma_{\mathbf{q}}^{(2)} - 1) & 0 \\ 0 & 0 & 6J_1 - BS^{-1} + 6J_2(\gamma_{\mathbf{q}}^{(2)} - 1) \end{pmatrix}, \quad (\text{S23})$$

and

$$H_{12}^0(\mathbf{q}) = S \begin{pmatrix} 0 & 0 & -3J_1\gamma_{-\mathbf{q}}^{(1)} \\ 0 & 0 & -3J_1\gamma_{\mathbf{q}}^{(1)} \\ -3J_1\gamma_{\mathbf{q}}^{(1)} & -3J_1\gamma_{-\mathbf{q}}^{(1)} & 0 \end{pmatrix}. \quad (\text{S24})$$

The corresponding structure factors are

$$\gamma_{\mathbf{q}}^{(1)} = \frac{1}{3} \left[e^{i\mathbf{q} \cdot \mathbf{a}_1} + e^{i\mathbf{q} \cdot \mathbf{a}_2} + e^{-i\mathbf{q} \cdot (\mathbf{a}_1 + \mathbf{a}_2)} \right], \quad (\text{S25})$$

$$\gamma_{\mathbf{q}}^{(2)} = \frac{1}{3} [\cos(\mathbf{q} \cdot (2\mathbf{a}_1 + \mathbf{a}_2)) + \cos(\mathbf{q} \cdot (\mathbf{a}_1 - \mathbf{a}_2)) + \cos(\mathbf{q} \cdot (\mathbf{a}_1 + 2\mathbf{a}_2))]. \quad (\text{S26})$$

The remaining blocks are fixed by the bosonic Bogoliubov structure,

$$H_{21}^0(\mathbf{q}) = [H_{12}^0(\mathbf{q})]^\dagger, \quad H_{22}^0(\mathbf{q}) = [H_{11}^0(-\mathbf{q})]^T. \quad (\text{S27})$$

The excitation spectrum of the LSWT Hamiltonian in Eq. (S21) contains two quadratic gapless modes at $\mathbf{q} = 0$, reflecting an accidental degeneracy of the classical UUD state. To lift this accidental degeneracy and open a gap in the spectrum, we include higher-order corrections in the $1/S$ expansion, which arise from interactions among the H-P bosons:

$$\mathcal{H}_{\text{int}} = \mathcal{H}^{(4)} + \mathcal{O}(S^{-1}), \quad (\text{S28})$$

where $\mathcal{H}^{(4)}$ denotes the quartic terms in the H-P boson operators. Cubic terms are absent because of the collinear nature of the UUD phase and the $U(1)$ symmetry of the model Hamiltonian [9].

To compute the correction to the single-magnon energies, we normal order the quartic interaction $\mathcal{H}^{(4)}$ in Eq. (S28). We distinguish onsite contractions from bond-weighted contractions. The onsite bilinears are defined as

$$\rho^\alpha = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \langle a_{\alpha, \mathbf{q}}^\dagger a_{\alpha, \mathbf{q}} \rangle, \quad \delta^\alpha = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \langle a_{\alpha, \mathbf{q}} a_{\alpha, -\mathbf{q}} \rangle. \quad (\text{S29})$$

For contractions associated with nearest-neighbor bonds ($\alpha \neq \beta$), we define

$$\xi^{\alpha\beta} = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \gamma_{\mathbf{q}}^{(1)} \langle a_{\alpha, \mathbf{q}}^\dagger a_{\beta, \mathbf{q}} \rangle, \quad (\text{S30})$$

$$\zeta^{\alpha\beta} = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \gamma_{\mathbf{q}}^{(1)} \langle a_{\alpha, \mathbf{q}} a_{\beta, -\mathbf{q}} \rangle. \quad (\text{S31})$$

For same-sublattice second-neighbor bonds ($\alpha = \beta$), we define analogously

$$\xi_2^{\alpha\alpha} = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \gamma_{\mathbf{q}}^{(2)} \langle a_{\alpha, \mathbf{q}}^\dagger a_{\alpha, \mathbf{q}} \rangle, \quad (\text{S32})$$

$$\zeta_2^{\alpha\alpha} = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \gamma_{\mathbf{q}}^{(2)} \langle a_{\alpha, \mathbf{q}} a_{\alpha, -\mathbf{q}} \rangle. \quad (\text{S33})$$

N_u is the number of magnetic unit cells.

Normal ordering generates an effective quadratic contribution, which we denote by $\mathcal{H}_{\text{NO}}$. This correction can be written in the Bogoliubov form

$$\mathcal{H}_{\text{NO}} = \frac{1}{2} \sum_{\mathbf{q} \in \text{MBZ}} ((\mathbf{a}_{\mathbf{q}}^\dagger)^T \ (\mathbf{a}_{-\mathbf{q}})^T) H_{\text{NO}}^{(2)}(\mathbf{q}) \begin{pmatrix} \mathbf{a}_{\mathbf{q}} \\ \mathbf{a}_{-\mathbf{q}}^\dagger \end{pmatrix}, \quad H_{\text{NO}}^{(2)}(\mathbf{q}) \equiv \begin{pmatrix} H_{11}^{\text{NO}}(\mathbf{q}) & H_{12}^{\text{NO}}(\mathbf{q}) \\ H_{21}^{\text{NO}}(\mathbf{q}) & H_{22}^{\text{NO}}(\mathbf{q}) \end{pmatrix}. \quad (\text{S34})$$

The block matrices are given by

$$H_{11}^{\text{NO}}(\mathbf{q}) = \begin{pmatrix} \mu^A + \mu_{J_2}^A + t_{J_2}^{AA} \gamma_{\mathbf{q}}^{(2)} & [t^{AB}]^* \gamma_{\mathbf{q}}^{(1)} & t^{CA} \gamma_{-\mathbf{q}}^{(1)} \\ t^{AB} \gamma_{-\mathbf{q}}^{(1)} & \mu^B + \mu_{J_2}^B + t_{J_2}^{BB} \gamma_{\mathbf{q}}^{(2)} & [t^{BC}]^* \gamma_{\mathbf{q}}^{(1)} \\ [t^{CA}]^* \gamma_{\mathbf{q}}^{(1)} & t^{BC} \gamma_{-\mathbf{q}}^{(1)} & \mu^C + \mu_{J_2}^C + t_{J_2}^{CC} \gamma_{\mathbf{q}}^{(2)} \end{pmatrix}, \quad (\text{S35})$$

and

$$H_{12}^{\text{NO}}(\mathbf{q}) = \begin{pmatrix} \Gamma^A + \Gamma_{J_2}^A + g_{J_2}^{AA} \gamma_{\mathbf{q}}^{(2)} & g^{AB} \gamma_{\mathbf{q}}^{(1)} & g^{CA} \gamma_{-\mathbf{q}}^{(1)} \\ g^{AB} \gamma_{-\mathbf{q}}^{(1)} & \Gamma^B + \Gamma_{J_2}^B + g_{J_2}^{BB} \gamma_{\mathbf{q}}^{(2)} & g^{BC} \gamma_{\mathbf{q}}^{(1)} \\ g^{CA} \gamma_{\mathbf{q}}^{(1)} & g^{BC} \gamma_{-\mathbf{q}}^{(1)} & \Gamma^C + \Gamma_{J_2}^C + g_{J_2}^{CC} \gamma_{\mathbf{q}}^{(2)} \end{pmatrix}. \quad (\text{S36})$$

As in the LSWT Hamiltonian, the remaining blocks are fixed by the bosonic Bogoliubov structure,

$$H_{21}^{\text{NO}}(\mathbf{q}) = [H_{12}^{\text{NO}}(\mathbf{q})]^\dagger, \quad H_{22}^{\text{NO}}(\mathbf{q}) = [H_{11}^{\text{NO}}(-\mathbf{q})]^T. \quad (\text{S37})$$

The coefficients are defined as

$$\begin{aligned} \mu^A &= 3J_1(\rho^B - \rho^C - \text{Re } \xi^{AB} + \text{Re } \zeta^{CA}), \\ \mu^B &= 3J_1(\rho^A - \rho^C - \text{Re } \xi^{AB} + \text{Re } \zeta^{BC}), \\ \mu^C &= 3J_1(-\rho^A - \rho^B + \text{Re } \zeta^{BC} + \text{Re } \zeta^{CA}), \\ t^{AB} &= 3J_1(\xi^{AB} - \tfrac{1}{2}(\rho^A + \rho^B)), \\ t^{BC} &= 3J_1(-\xi^{BC} + \tfrac{1}{4}([\delta^B]^* + \delta^C)), \\ t^{CA} &= 3J_1(-\xi^{CA} + \tfrac{1}{4}([\delta^C]^* + \delta^A)), \\ \Gamma^A &= \tfrac{3J_1}{2}(\xi^{CA} - \zeta^{AB}), \\ \Gamma^B &= \tfrac{3J_1}{2}([\xi^{BC}]^* - \zeta^{AB}), \\ \Gamma^C &= \tfrac{3J_1}{2}([\xi^{BC}]^* - \xi^{CA}), \\ g^{AB} &= 3J_1(\zeta^{AB} - \tfrac{1}{4}(\delta^A + \delta^B)), \\ g^{BC} &= 3J_1(-\zeta^{BC} + \tfrac{1}{2}(\rho^B + \rho^C)), \\ g^{CA} &= 3J_1(-\zeta^{CA} + \tfrac{1}{2}(\rho^C + \rho^A)), \end{aligned} \quad (\text{S38})$$

and the J_2 corrections are

$$\mu_{J_2}^\alpha = 6J_2(\rho^\alpha - \text{Re } \xi_2^{\alpha\alpha}), \quad (\text{S39})$$

$$t_{J_2}^{\alpha\alpha} = 6J_2([\xi_2^{\alpha\alpha}]^* - \rho^\alpha), \quad (\text{S40})$$

$$\Gamma_{J_2}^\alpha = -\tfrac{3J_2}{2}[\zeta_2^{\alpha\alpha}]^*, \quad (\text{S41})$$

$$g_{J_2}^{\alpha\alpha} = 3J_2[\zeta_2^{\alpha\alpha} - \delta^\alpha]^*. \quad (\text{S42})$$

At order $1/S$, the leading correction to the LSWT spectrum is obtained by treating $H_{\text{NO}}^{(2)}(\mathbf{q})$ perturbatively. We refer to this as the *one-loop correction*. In this treatment, the bilinears in Eqs. (S29)–(S33) are evaluated using the LSWT vacuum. Let $T_{\mathbf{q}}$ be the Bogoliubov transformation that diagonalizes the LSWT quadratic Hamiltonian,

$$H_0^{(2)}(\mathbf{q}) \xrightarrow{T_{\mathbf{q}}} \text{diag}(\varepsilon_{1,\mathbf{q}}^0, \varepsilon_{2,\mathbf{q}}^0, \varepsilon_{3,\mathbf{q}}^0, -\varepsilon_{1,-\mathbf{q}}^0, -\varepsilon_{2,-\mathbf{q}}^0, -\varepsilon_{3,-\mathbf{q}}^0). \quad (\text{S43})$$

We write

$$\begin{pmatrix} \mathbf{a}_{\mathbf{q}} \\ \mathbf{a}_{-\mathbf{q}}^\dagger \end{pmatrix} = T_{\mathbf{q}} \begin{pmatrix} \mathbf{b}_{\mathbf{q}} \\ \mathbf{b}_{-\mathbf{q}}^\dagger \end{pmatrix}, \quad T_{\mathbf{q}} = \begin{pmatrix} U_{\mathbf{q}} & V_{\mathbf{q}} \\ V_{\mathbf{q}} & U_{\mathbf{q}} \end{pmatrix}. \quad (\text{S44})$$

The onsite LSWT contractions are then

$$\rho^\alpha = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \sum_{n=1}^3 [V_{\mathbf{q}}]_{\alpha n}^* [V_{\mathbf{q}}]_{\alpha n}, \quad (\text{S45})$$

$$\delta^\alpha = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \sum_{n=1}^3 [U_{\mathbf{q}}]_{\alpha n} [V_{\mathbf{q}}]_{\alpha n}^*. \quad (\text{S46})$$

The bond-weighted LSWT contractions are

$$\xi^{\alpha\beta} = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \gamma_{\mathbf{q}}^{(1)} \sum_{n=1}^3 [V_{\mathbf{q}}]_{\alpha n}^* [V_{\mathbf{q}}]_{\beta n}, \quad (\text{S47})$$

$$\zeta^{\alpha\beta} = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \gamma_{\mathbf{q}}^{(1)} \sum_{n=1}^3 [U_{\mathbf{q}}]_{\alpha n} [V_{\mathbf{q}}]_{\beta n}^*, \quad (\text{S48})$$

$$\xi_2^{\alpha\alpha} = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \gamma_{\mathbf{q}}^{(2)} \sum_{n=1}^3 [V_{\mathbf{q}}]_{\alpha n}^* [V_{\mathbf{q}}]_{\alpha n}, \quad (\text{S49})$$

$$\zeta_2^{\alpha\alpha} = \frac{1}{N_u} \sum_{\mathbf{q} \in \text{MBZ}} \gamma_{\mathbf{q}}^{(2)} \sum_{n=1}^3 [U_{\mathbf{q}}]_{\alpha n} [V_{\mathbf{q}}]_{\alpha n}^*. \quad (\text{S50})$$

Substituting these LSWT contractions into $H_{\text{NO}}^{(2)}(\mathbf{q})$, the one-loop correction to the n -th single-magnon energy is given by the diagonal matrix element

$$\delta\varepsilon_{n,\mathbf{q}}^{\text{1L}} = \left[T_{\mathbf{q}}^\dagger H_{\text{NO}}^{(2)}(\mathbf{q}) T_{\mathbf{q}} \right]_{nn}, \quad n = 1, 2, 3. \quad (\text{S51})$$

The one-loop spectrum is therefore

$$\varepsilon_{n,\mathbf{q}}^{\text{1L}} = \varepsilon_{n,\mathbf{q}}^0 + \delta\varepsilon_{n,\mathbf{q}}^{\text{1L}}. \quad (\text{S52})$$

In this perturbative scheme, both $T_{\mathbf{q}}$ and the contractions ρ^α , δ^α , $\xi^{\alpha\beta}$, $\zeta^{\alpha\beta}$, $\xi_2^{\alpha\alpha}$, and $\zeta_2^{\alpha\alpha}$ are computed solely from the LSWT Hamiltonian $H_0^{(2)}(\mathbf{q})$.

In the *self-consistent one-loop theory*, by contrast, the normal-ordering correction is fed back into the quadratic Hamiltonian. Namely, one diagonalizes

$$H_{\text{SCOL}}^{(2)}(\mathbf{q}) = H_0^{(2)}(\mathbf{q}) + H_{\text{NO}}^{(2)}(\mathbf{q}), \quad (\text{S53})$$

where $H_{\text{NO}}^{(2)}(\mathbf{q})$ is evaluated using contractions obtained from the vacuum of $H_{\text{SCOL}}^{(2)}(\mathbf{q})$ itself. The procedure is iterated according to

$$\{\rho^\alpha, \delta^\alpha, \xi^{\alpha\beta}, \zeta^{\alpha\beta}, \xi_2^{\alpha\alpha}, \zeta_2^{\alpha\alpha}\} \longrightarrow H_{\text{NO}}^{(2)}(\mathbf{q}) \longrightarrow H_{\text{SCOL}}^{(2)}(\mathbf{q}) \longrightarrow T_{\mathbf{q}}^{\text{SC}} \longrightarrow \{\rho^\alpha, \delta^\alpha, \xi^{\alpha\beta}, \zeta^{\alpha\beta}, \xi_2^{\alpha\alpha}, \zeta_2^{\alpha\alpha}\}, \quad (\text{S54})$$

until convergence is reached. The resulting excitation spectrum is obtained from the positive Bogoliubov eigenvalues of the converged matrix $H_{\text{SC}}^{(2)}(\mathbf{q})$. Thus, the one-loop theory treats $H_{\text{NO}}^{(2)}$ as a leading $1/S$ perturbative correction evaluated in the LSWT vacuum, whereas the self-consistent theory incorporates $H_{\text{NO}}^{(2)}$ directly into the quadratic Hamiltonian and updates the bosonic contractions iteratively.

To determine the plateau phase boundaries, we compute the renormalized single-magnon spectrum away from the classical stability field $B_{\text{cl}} = 3J_1S$. In doing so, we assume that the bosonic bilinears are only weakly dependent on the external magnetic field within the plateau. We therefore evaluate these bilinears at B_{cl} and keep them fixed when varying B .

For a given field B , the LSWT contribution is obtained from the field-dependent quadratic matrix $H_0^{(2)}(\mathbf{q}; B)$, while the normal-ordering correction $H_{\text{NO}}^{(2)}(\mathbf{q})$ is constructed using the bilinears evaluated at B_{cl} . In the one-loop approach, the magnon energies are computed by treating $H_{\text{NO}}^{(2)}(\mathbf{q})$ as the leading $1/S$ correction to the LSWT spectrum at the corresponding field B . In the self-consistent approach, we instead diagonalize the renormalized quadratic matrix

$$H_{\text{SCOL}}^{(2)}(\mathbf{q}; B) = H_0^{(2)}(\mathbf{q}; B) + H_{\text{NO}}^{(2)}(\mathbf{q}), \quad (\text{S55})$$

where $H_{\text{NO}}^{(2)}(\mathbf{q})$ is fixed by the self-consistent bilinears obtained at B_{cl} .

The lower and upper critical fields, h_{c1} and h_{c2} , are determined by tracking the minimum positive magnon gap as a function of B . The plateau remains stable as long as all renormalized magnon energies are positive. The phase boundaries are identified by the first closing of the gap,

$$\min_{n,\mathbf{q}} \varepsilon_{n,\mathbf{q}}(B = h_{c1}) = 0, \quad \min_{n,\mathbf{q}} \varepsilon_{n,\mathbf{q}}(B = h_{c2}) = 0, \quad (\text{S56})$$

on the low-field and high-field sides of the plateau, respectively. Numerically, we locate these gap-closing fields using bisection. In this way, the $1/S$ correction shifts the plateau boundaries away from the classical value B_{cl} , thereby producing a finite field window for the magnetization plateau.

IV. HIGH-FIELD ANALYSIS OF THE MAGNON CONDENSATION FOR $J_2/J_1 > 1/8$

To understand the phase diagram of the spin-1/2 model right below the saturation field, we use a hard-core boson representation of spin operators, $S^z = 1/2 - a^+a$, $S^+ = a$, $S^- = a^+$, and perform the magnon Bose-Einstein condensation analysis [10, 11]. The analysis begins with the Hamiltonian of magnons in the fully polarized state,

$$H = \sum_{\mathbf{k}} \epsilon(\mathbf{k}) a_{\mathbf{k}}^+ a_{\mathbf{k}} + \frac{1}{2N} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} V(\mathbf{q}) a_{\mathbf{k}+\mathbf{q}}^+ a_{\mathbf{k}'-\mathbf{q}}^+ a_{\mathbf{k}'} a_{\mathbf{k}}, \quad (\text{S57})$$

where the two-particle interaction

$$V(\mathbf{q}) = 2U + 6J_1\gamma_1(\mathbf{q}) + 6J_2\gamma_2(\mathbf{q}) \quad (\text{S58})$$

originates entirely from the $S^z - S^z$ part of the spin exchange interaction and the hard-core constraint term $U \sum_{\mathbf{r}} \hat{n}_{\mathbf{r}}(\hat{n}_{\mathbf{r}} - 1)$ which restricts the boson on-site occupation number $\hat{n}_{\mathbf{r}}$ to eigenvalues 0 and 1 only, for every $\mathbf{r}$, in the limit $U \rightarrow \infty$. Here and below, $\gamma_{1,2}(\mathbf{q})$ are related to (S8) via $\gamma_1(\mathbf{q}) = (\gamma_{\mathbf{q}} + \gamma_{-\mathbf{q}})/2 = 1/3 \sum_j \cos(\mathbf{q} \cdot \boldsymbol{\delta}_j)$, $\gamma_2(\mathbf{q}) = \gamma_{\mathbf{q}}^{(2)} = 1/3 \sum_j \cos(\mathbf{q} \cdot \mathbf{e}_j)$. The next-nearest neighbor vectors $\mathbf{e}_j$ of the lattice are related to $\mathbf{a}_j$ in Eq. (S8) by $\mathbf{e}_1 = \mathbf{a}_1, \mathbf{e}_2 = \mathbf{a}_3, \mathbf{e}_3 = \mathbf{a}_2$.

Magnon dispersion in the fully polarized state is given by

$$\epsilon(\mathbf{k}) = 3J_1\gamma_1(\mathbf{k}) + 3J_2\gamma_2(\mathbf{k}) + J_1 + J_2 + \tilde{\mu}, \quad (\text{S59})$$

where $\tilde{\mu} = B - B_{\text{sat}}$, $B_{\text{sat}} = 4(J_1 + J_2)$, measures the distance to the saturation field. The minimum of the dispersion, given by $\tilde{\mu}$, is reached at the three M-points of the hexagonal Brillouin zone of the triangular lattice: $M_1 = (0, 2\pi/\sqrt{3})$, $M_2 = (\pi, \pi/\sqrt{3})$, $M_3 = (-\pi, \pi/\sqrt{3})$. These points are located at the mid-faces of the BZ and, therefore, vectors $2M_{i=1,2,3}$ are the reciprocal lattice vectors. The presence of the three degenerate minima opens up the competition between the *single-Q* and *multi-Q* condensation scenarios at $\mu = 0$, when magnons become gapless. The single-Q case corresponds to the spontaneous condensation of magnons at *one* of the three possible minima M_i . The multi-Q case occurs when the magnon experiences coherent condensation at all three M points.

To describe the condensation, we parameterize the magnon operator in terms of c -number condensates ϕ_i at momentum M_i [12]

$$a_{\mathbf{k}} = \sqrt{N}\phi_1\delta_{\mathbf{k},M_1} + \sqrt{N}\phi_2\delta_{\mathbf{k},M_2} + \sqrt{N}\phi_3\delta_{\mathbf{k},M_3} + \tilde{a}_{\mathbf{k}}, \quad (\text{S60})$$

where operator $\tilde{a}_{\mathbf{k}}$ describes non-condensed magnons, and use $\bar{\phi}_i$ for the complex conjugate of ϕ_i .

The competition between the single-Q and multi-Q condensations is captured by the effective Ginzburg-Landau functional

$$\begin{aligned} \frac{E}{N} = & \tilde{\mu}(|\phi_1|^2 + |\phi_2|^2 + |\phi_3|^2) + \frac{1}{2}\Gamma_1(|\phi_1|^4 + |\phi_2|^4 + |\phi_3|^4) + \Gamma_2(|\phi_1|^2|\phi_2|^2 + |\phi_1|^2|\phi_3|^2 + |\phi_2|^2|\phi_3|^2) \\ & + \Gamma_u(\bar{\phi}_1^2\phi_2^2 + \bar{\phi}_1^2\phi_3^2 + \bar{\phi}_2^2\phi_3^2 + \text{h.c.}) \end{aligned} \quad (\text{S61})$$

where $\Gamma_{1,2,u}$ are obtained from the fully renormalized four-point vertex function $\Gamma(\mathbf{k}_1, \mathbf{k}_2; \mathbf{q})$. Namely,

$$\Gamma_1 = \Gamma(M_1, M_1; \mathbf{q} = 0); \Gamma_2 = \Gamma(M_1, M_2; \mathbf{q} = 0) + \Gamma(M_1, M_2; \mathbf{q} = M_3); \Gamma_u = \Gamma(M_1, M_1; \mathbf{q} = M_3). \quad (\text{S62})$$

A simple analysis of (S61) shows that $\Gamma_1 < \Gamma_2$ makes it energetically favorable to condense at a single M-point. This situation corresponds to the *stripe* phase. In the opposite case of $\Gamma_1 > \Gamma_2$, the condensate develops coherently at all three M points, resulting in the coplanar $\bar{V}$ state [12].

The renormalized vertex function obeys the Bethe-Salpeter equation

$$\Gamma(\mathbf{k}_1, \mathbf{k}_2; \mathbf{q}) = V(\mathbf{q}) - \frac{1}{N} \sum_{\mathbf{q}' \in \text{BZ}} \frac{V(\mathbf{q} - \mathbf{q}')\Gamma(\mathbf{k}_1, \mathbf{k}_2; \mathbf{q}')}{\epsilon(\mathbf{k}_1 + \mathbf{q}') + \epsilon(\mathbf{k}_2 - \mathbf{q}') + \mu} \quad (\text{S63})$$

where $\mu = 2\tilde{\mu}$. In the following, we set $J_1 = 1$ and $J_2 = j$.

(a) Consider the case $\mathbf{k}_1 = \mathbf{k}_2 = M_1$. Then

$$\begin{aligned} D_a(\mathbf{q}) = & \epsilon(M_1 + \mathbf{q}) + \epsilon(M_1 - \mathbf{q}) + \mu = 2(1 + j) + \cos(\mathbf{q} \cdot \boldsymbol{\delta}_1) - \cos(\mathbf{q} \cdot \boldsymbol{\delta}_2) - \cos(\mathbf{q} \cdot \boldsymbol{\delta}_3) \\ & - j[\cos(\mathbf{q} \cdot \mathbf{e}_1) - \cos(\mathbf{q} \cdot \mathbf{e}_2) + \cos(\mathbf{q} \cdot \mathbf{e}_3)] + \mu = D_a(-\mathbf{q}) \end{aligned} \quad (\text{S64})$$

In addition to being even function of $\mathbf{q}$, the denominator is also invariant under the inversion $(q_x, q_y) \rightarrow (-q_x, q_y)$, which results in invariance under $\delta_2 \leftrightarrow \delta_3$ and $\mathbf{e}_1 \leftrightarrow \mathbf{e}_3$. The interaction potential $V(\mathbf{q})$ shares these properties. This suggests the following ansatz for the vertex function

$$\Gamma(\mathbf{k}_1, \mathbf{k}_2; \mathbf{q}) = \sum_{i=0}^4 \eta_i \psi_i(\mathbf{q}) \quad (\text{S65})$$

with the basis vectors

$$\begin{aligned} \psi_0(\mathbf{q}) &= 1; \psi_1(\mathbf{q}) = \sqrt{2} \cos(\mathbf{q} \cdot \delta_1); \psi_2(\mathbf{q}) = \cos(\mathbf{q} \cdot \delta_2) + \cos(\mathbf{q} \cdot \delta_3); \\ \psi_3(\mathbf{q}) &= \cos(\mathbf{q} \cdot \mathbf{e}_1) + \cos(\mathbf{q} \cdot \mathbf{e}_3); \psi_4(\mathbf{q}) = \sqrt{2} \cos(\mathbf{q} \cdot \mathbf{e}_2). \end{aligned} \quad (\text{S66})$$

These functions form an orthonormal basis, $1/N \sum_{\mathbf{q} \in \text{BZ}} \psi_i(\mathbf{q}) \psi_j(\mathbf{q}) = \delta_{ij}$. Not surprisingly, the interaction potential (S58) can be written in this basis as well, $V(\mathbf{q}) = \sum_{i=0}^4 v_i \psi_i(\mathbf{q})$, where $v_i = (2U, \sqrt{2}, 2, 2j, \sqrt{2}j)$.

From here, we reduce (S63) to the matrix equation by multiplying its both sides with $\psi_j(\mathbf{q})$ and integrating over $\mathbf{q}$, for $j = 0 \dots 4$. Since $\psi_0 = 1$, the first of these equations is obtained by integrating both sides of (S63) over $\mathbf{q}$ and using $1/N \sum_{\mathbf{q}} \psi_n(\mathbf{q}) = 0$ for $n = 1, 2, 3, 4$. We obtain $\eta_0/(2U) = 1 - 1/N \sum_{\mathbf{q}'} \Gamma(\mathbf{k}_1, \mathbf{k}_2; \mathbf{q}')/D_a(\mathbf{q}')$ and note that in the hard-core $U \rightarrow \infty$ limit the left-hand side vanishes. Upon introducing

$$\tau_{ij} = \tau_{ji} = \frac{1}{N} \sum_{\mathbf{q} \in \text{BZ}} \frac{\psi_i(\mathbf{q}) \psi_j(\mathbf{q})}{D_a(\mathbf{q})}, \quad (\text{S67})$$

this equation can be compactly written as $\sum_{i=0}^4 \tau_{0i} \eta_i = 1$. Next, multiply (S63) by $\psi_{j=1,2,3,4}$ and integrate over $\mathbf{q}$. Using easily derived identities

$$\begin{aligned} \frac{1}{N} \sum_{\mathbf{q} \in \text{BZ}} \psi_{j=1,4}(\mathbf{q}) \psi_{i=1,4}(\mathbf{q} - \mathbf{q}') &= \frac{1}{\sqrt{2}} \psi_{j=1,4}(\mathbf{q}') \delta_{ji}, \\ \frac{1}{N} \sum_{\mathbf{q} \in \text{BZ}} \psi_{j=2,3}(\mathbf{q}) \psi_{i=2,3}(\mathbf{q} - \mathbf{q}') &= \frac{1}{2} \psi_{j=2,3}(\mathbf{q}') \delta_{ji} \end{aligned} \quad (\text{S68})$$

we obtain

$$\eta_j + \frac{1}{N} \sum_{\mathbf{q}' \in \text{BZ}} \sum_{i=0}^4 \eta_i \frac{\psi_i(\mathbf{q}')}{D_a(\mathbf{q}')} \left[\frac{v_1}{\sqrt{2}} \psi_1(\mathbf{q}') \delta_{j1} + \frac{v_2}{2} \psi_2(\mathbf{q}') \delta_{j2} + \frac{v_3}{2} \psi_3(\mathbf{q}') \delta_{j3} + \frac{v_4}{\sqrt{2}} \psi_4(\mathbf{q}') \delta_{j4} \right] = v_j \quad (\text{S69})$$

Altogether, (S63) turns into the matrix equation for vector $\vec{\eta} = (\eta_0, \eta_1, \eta_2, \eta_3, \eta_4)^t$

$$\begin{pmatrix} \tau_{00} & \tau_{01} & \tau_{02} & \tau_{03} & \tau_{04} \\ \tau_{10} & 1 + \tau_{11} & \tau_{12} & \tau_{13} & \tau_{14} \\ \tau_{20} & \tau_{12} & 1 + \tau_{22} & \tau_{23} & \tau_{24} \\ j\tau_{30} & j\tau_{13} & j\tau_{23} & 1 + j\tau_{33} & j\tau_{34} \\ j\tau_{40} & j\tau_{14} & j\tau_{24} & j\tau_{34} & 1 + j\tau_{44} \end{pmatrix} \begin{pmatrix} \eta_0 \\ \eta_1 \\ \eta_2 \\ \eta_3 \\ \eta_4 \end{pmatrix} = \begin{pmatrix} 1 \\ \sqrt{2} \\ 2 \\ 2j \\ \sqrt{2}j \end{pmatrix} \quad (\text{S70})$$

Once $\vec{\eta}$ is found, $\Gamma_{1,u}$ follow via

$$\Gamma_1 = \sum_{i=0}^4 \eta_i \psi_i(\mathbf{q} = 0) = \vec{\psi}(\mathbf{q} = 0) \cdot \vec{\eta}; \quad \Gamma_u = \sum_{i=0}^4 \eta_i \psi_i(\mathbf{q} = M_3) = \vec{\psi}(\mathbf{q} = M_3) \cdot \vec{\eta}. \quad (\text{S71})$$

Here $\vec{\psi} = (\psi_0, \psi_1, \psi_2, \psi_3, \psi_4)$ is the vector of the basis functions (S66), and we denote the vector on the right-hand side as $\vec{w} = (1, \sqrt{2}, 2, 2j, \sqrt{2}j)$.

Each of τ_{ij} diverges logarithmically in the limit $\mu \rightarrow 0$. The divergence comes from the neighborhood of points $\mathbf{q}' = 0, M_2, M_3$ in $D_a(\mathbf{q}')$. The diverging contribution to τ_{ij} is found to be

$$\tilde{\tau}_{ij} = (\psi_i(0) \psi_j(0) + 2\psi_i(M_3) \psi_j(M_3)) L_a(j, \mu), \quad (\text{S72})$$

where

$$L_a(j, \mu) = \frac{1}{4\pi \sqrt{4j - (3j - 1)^2}} \ln\left(\frac{1}{\mu}\right). \quad (\text{S73})$$

We then split $\tau_{ij} = \tilde{\tau}_{ij} + \tau_{ij}^0$, where $\tau_{ij}^0 = \tau_{ij} - \tilde{\tau}_{ij}$ is the regular contribution which is evaluated numerically by calculating the difference of the integral τ_{ij} (S67) and (S72) in the $\mu \rightarrow 0$ limit. Practically, we find τ_{ij}^0 to reach a μ -independent value for $\mu \sim 10^{-6} - 10^{-8}$.

Finally, we solve (S70) for $\vec{\eta}$ by inverting the matrix with the help of Mathematica and expanding the result in inverse powers of diverging $L_a(j, \mu \rightarrow 0)$ to find

$$\Gamma_1(j, \mu) = \frac{1}{L_a(j, \mu)} + \frac{c_a(j)}{L_a^2(j, \mu)} + \dots; \quad \Gamma_u(j, \mu) = \frac{c_u(j)}{L_a^2(j, \mu)} + \dots \quad (\text{S74})$$

Here, j -dependent coefficients $c_{a/u}(j)$ are determined by the *regular* τ_{ij}^0 contributions. We find $c_a(1/3) \approx -0.085$ and $c_u(1/3) \approx -0.011$. The fact that $\Gamma_u \ll \Gamma_1$ is consistent with the previous large- S calculation [12].

(b) Now consider $\mathbf{k}_1 = M_1, \mathbf{k}_2 = M_2$. Then

$$D_b(\mathbf{q}) = \epsilon(M_1 + \mathbf{q}) + \epsilon(M_2 - \mathbf{q}) + \mu = 2(1 + j) - 2\cos(\mathbf{q} \cdot \boldsymbol{\delta}_3) - 2j\cos(\mathbf{q} \cdot \mathbf{e}_1) + \mu = D_b(-\mathbf{q}) \quad (\text{S75})$$

This suggests a slightly different set of basis functions

$$\begin{aligned} \chi_0(\mathbf{q}) &= 1; \chi_1(\mathbf{q}) = \cos(\mathbf{q} \cdot \boldsymbol{\delta}_1) + \cos(\mathbf{q} \cdot \boldsymbol{\delta}_2); \chi_2(\mathbf{q}) = \sqrt{2}\cos(\mathbf{q} \cdot \boldsymbol{\delta}_3) \\ \chi_3(\mathbf{q}) &= \sqrt{2}\cos(\mathbf{q} \cdot \mathbf{e}_1); \chi_4(\mathbf{q}) = \cos(\mathbf{q} \cdot \mathbf{e}_2) + \cos(\mathbf{q} \cdot \mathbf{e}_3). \end{aligned} \quad (\text{S76})$$

We expand $\Gamma(\mathbf{k}_1, \mathbf{k}_2; \mathbf{q}) = \sum_{i=0}^4 \eta'_i \chi_i(\mathbf{q})$ and $V(\mathbf{q}) = \sum_{i=0}^4 v'_i \chi_i(\mathbf{q})$, where $v'_i = (2U, 2, \sqrt{2}, \sqrt{2}j, 2j)$. It is straightforward to derive the matrix equation for $\vec{\eta}'$, which is formulated in terms of

$$s_{ij} = \frac{1}{N} \sum_{\mathbf{q} \in \text{BZ}} \frac{\chi_i(\mathbf{q})\chi_j(\mathbf{q})}{D_b(\mathbf{q})}, \quad (\text{S77})$$

and has vector $\vec{w}' = (1, 2, \sqrt{2}, \sqrt{2}j, 2j)$ in the right-hand side. Solving it gives

$$\Gamma_2 = \sum_{i=0}^4 \eta'_i (\chi_i(\mathbf{q} = 0) + \chi_i(\mathbf{q} = M_3)). \quad (\text{S78})$$

In distinction from the previous case, here s_{ij} diverges logarithmically when $\mathbf{q}' = 0$ or M_3 . The diverging contribution $\tilde{s}_{ij}$ reads

$$\tilde{s}_{ij} = (\chi_i(0)\chi_j(0) + \chi_i(M_3)\chi_j(M_3))L_b(j, \mu), \text{ where } L_b(j, \mu) = \frac{1}{4\pi\sqrt{4j}} \ln\left(\frac{1}{\mu}\right). \quad (\text{S79})$$

At the end of the day, we find that in the $\mu \rightarrow 0$ limit

$$\Gamma_2(j, \mu) = \frac{1}{L_b(j, \mu)} + \frac{c_b(j)}{L_b^2(j, \mu)} + \dots \text{ with } c_b(1/3) \approx -0.092. \quad (\text{S80})$$

Therefore,

$$\Gamma_1(j, \mu) - \Gamma_2(j, \mu) = 4\pi \frac{\sqrt{4j - (3j - 1)^2} - \sqrt{4j}}{\ln(1/\mu)} + O\left(\frac{1}{\ln^2(1/\mu)}\right) \quad (\text{S81})$$

That is, for every $j \neq 1/3$, $\Gamma_1 < \Gamma_2$ because the coefficient of the leading, $1/\ln(1/\mu)$, contribution is negative. The subleading $1/\ln^2(1/\mu)$ terms can not change this outcome in the $\mu \rightarrow 0$ limit. This means the *stripe* magnetic order.

The only exception to this occurs at $j = 1/3$, when the leading contributions cancel out. At that point, $1/\ln^2(1/\mu)$ terms ‘flip’ the outcome and lead to $\Gamma_1 > \Gamma_2$. At this special point, it is the $\bar{V}$ phase that is stabilized just below the saturation field. Our result, *exact* for the spin $S = 1/2$ $J_1 - J_2$ triangular lattice antiferromagnet, confirms previous semi-classical, $S \gg 1$, analysis where the direct transition from the polarized to $\bar{V}$ state at $J_2 = J_1/3$ was first suggested [12].

[1] A. L. Chernyshev and O. A. Starykh, Roller Coaster in a Flatland: Magnetoresistivity in Eu-Intercalated Graphite, [Phys. Rev. X](#) **12**, 021010 (2022).

- [2] V. N. Krivoruchko and D. A. Yablonskii, On the Problem of Calculating the Bogolyubov U-V Transformation Coefficients, *physica status solidi (b)* **103**, K41 (1981).
- [3] P. A. Maksimov, Z. Zhu, S. R. White, and A. L. Chernyshev, Anisotropic-exchange magnets on a triangular lattice: Spin waves, accidental degeneracies, and dual spin liquids, *Phys. Rev. X* **9**, 021017 (2019).
- [4] J. Alicea, A. V. Chubukov, and O. A. Starykh, Quantum Stabilization of the 1/3-Magnetization Plateau in Cs_2CuBr_4 , *Physical Review Letters* **102**, 137201 (2009).
- [5] Y. Kamiya, L. Ge, T. Hong, Y. Qiu, D. L. Quintero-Castro, Z. Lu, H. B. Cao, M. Matsuda, E. S. Choi, C. D. Batista, M. Mourigal, H. D. Zhou, and J. Ma, The nature of spin excitations in the one-third magnetization plateau phase of $\text{Ba}_3\text{CoSb}_2\text{O}_9$, *Nature Communications* **9**, 2666 (2018).
- [6] H. Zhang, *GitHub repository for semiclassical spin-wave analysis of triangular-lattice magnetization plateaux* (2026).
- [7] D. Dahlbom, H. Zhang, C. Miles, S. Quinn, A. Niraula, B. Thipe, M. Wilson, S. Matin, H. Mankad, S. Hahn, D. Pajerowski, S. Johnston, Z. Wang, H. Lane, Y. W. Li, X. Bai, M. Mourigal, C. D. Batista, and K. Barros, *Sunny.jl GitHub Repository* (2025).
- [8] D. Dahlbom, H. Zhang, C. Miles, S. Quinn, A. Niraula, B. Thipe, M. Wilson, S. Matin, H. Mankad, S. Hahn, D. Pajerowski, S. Johnston, Z. Wang, H. Lane, Y. W. Li, X. Bai, M. Mourigal, C. D. Batista, and K. Barros, Sunny.jl: A Julia Package for Spin Dynamics, *Journal of Open Source Software* **10**, 8138 (2025).
- [9] A. L. Chernyshev and M. E. Zhitomirsky, Spin Waves in a Triangular Lattice Antiferromagnet: Decays, Spectrum Renormalization, and Singularities, *Physical Review B* **79**, 144416 (2009).
- [10] T. Nikuni and H. Shiba, Hexagonal Antiferromagnets in Strong Magnetic Field: Mapping onto Bose Condensation of Low-Density Bose Gas, *Journal of the Physical Society of Japan* **64**, 3471 (1995).
- [11] V. Zapf, M. Jaime, and C. D. Batista, Bose-Einstein condensation in quantum magnets, *Rev. Mod. Phys.* **86**, 563 (2014).
- [12] M. Ye and A. V. Chubukov, Quantum phase transitions in the Heisenberg $J_1 - J_2$ triangular antiferromagnet in a magnetic field, *Phys. Rev. B* **95**, 014425 (2017).